

Consent**Thank you for taking the time to participate in this survey!**

Title of Project: Investigating pedagogical approaches for teaching quantitative and computational skills to non-math majors

Principal Investigator: Dr. Miguel Acevedo

Please read the information below carefully before you decide to participate in this research study. **Your participation is voluntary. You can decide not to participate or later decide to stop participating at any time without penalty or lose any benefits that would normally be expected.**

1. Purpose of the Study: The purpose of this research study is to investigate pedagogical approaches for teaching both computational and quantitative/statistical skills to non-math major undergraduate students.
2. What will you be asked to do: You will be asked to answer questions regarding your experience with R and classes with quantitative components.
3. Time Required: It will take about 15 minutes to participate in the research.
4. Research Benefits: There are no direct benefits to you for being in this study. There may be a benefit to others depending on the results of this study.
5. Research Risks: There are no risks or discomforts anticipated.
6. Statement of Confidentiality: Your participation in this research is confidential. Information collected about you will be stored in computers with security passwords or in locked filing cabinets. Only certain people have the legal right to review these research records, and they will protect the secrecy (confidentiality) of these records as much as the law allows. These people include the researchers for this study, certain University of Florida officials, and the Institutional Review Board (IRB; an IRB is a group of people who are responsible for looking after the rights and welfare of people taking part in research). Otherwise your research records will not be released without your permission unless required by law or a court order. The researchers will not share your name or other identifiable information about you if they publish, present, or share the results this research.
7. Who to contact if you have questions: Please contact Miguel Acevedo at (352) 846-0575 with questions or concerns about this study.
8. Voluntary Participation: Your decision to be in this research is voluntary. You do not have to do any study activities that you do not want to take part in. You can stop at any time. If you decide you want to stop participating in the research, you can let the research team know or call the Principal Investigator at any time at (352) 846-0575. If you choose not to take part, this will have no effect on you or your relationships with the University of

Florida. If you have any questions about your rights as a research subject, you can phone the Institutional Review Board at 352-273-9600.

Participation in the research implies that you have read the information in this form and consent to take part in the research. Please save a copy of this form for your records or future reference.

If you want to participate in this research study, select "I consent" below. If you do not want to participate, you may simply close this window or select "I do not consent" below.

- ☐ I consent, begin the study
- ☐ I do not consent, I do not wish to participate

Technology Access

What type of operating system (OS) did you primarily use for this class?

- ☐ Mac
- ☐ Windows
- ☐ Linux

Roughly how old is the computer you primarily used for this class?

- ☐ 0 - 2 years old
- ☐ 2 - 4 years old
- ☐ 4 - 6 years old
- ☐ 6+ years old

During this past semester, how often did problems with your primary laptop (e.g., battery life, slowness, connectivity issues, etc.) negatively impact your performance in this class?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Often
- ☐ Always

Qualitative assessment

Providing as much detail as you can, please describe how you would produce a graph for a lab report or presentation using whichever program or software you prefer if asked to do so today.

What challenges do you feel stand in your way for learning R?

What benefits do you see coming from using R or other programming languages?

R Proficiency

Answer the following questions about your experiences with R *before and after* this semester.

	Before this semester					After this semester				
	Very unlikely	Somewhat unlikely	Neutral	Somewhat likely	Very likely	Very unlikely	Somewhat unlikely	Neutral	Somewhat likely	
How likely would you be to use R to analyze data?	☐	☐	☐	☐	☐	☐	☐	☐	☐	

	Before this semester					After this semester			
	Very low anxiety	Low anxiety	Moderate anxiety	High anxiety	Very high anxiety	Very low anxiety	Low anxiety	Moderate anxiety	High anxiety
How would you rate your anxiety surrounding computer programming tasks (e.g., using R)?	☐	☐	☐	☐	☐	☐	☐	☐	☐

	Before this semester					After this semester			
	Very low anxiety	Low anxiety	Moderate anxiety	High anxiety	Very high anxiety	Very low anxiety	Low anxiety	Moderate anxiety	High anxiety
How would you rate your anxiety surrounding mathematical concepts and problems?	☐	☐	☐	☐	☐	☐	☐	☐	☐

How would you rate your overall proficiency in the following programs?

	Before this semester					After this semester	
	No proficiency	Slightly proficient	Moderately proficient	Very proficient	Extremely proficient	No proficiency	Slightly proficient
R	☐	☐	☐	☐	☐	☐	☐
Spreadsheets (e.g., Excel, GoogleSheets)	☐	☐	☐	☐	☐	☐	☐
Other coding languages (e.g., Python, C++, SAS, MATLAB, etc.)	☐	☐	☐	☐	☐	☐	☐
Graphical user interface-based (GUI) statistical software programs (e.g., JMP, Minitab, SPSS, etc.)	☐	☐	☐	☐	☐	☐	☐

How would you rate your proficiency in the following skills **in R**?

	Before this semester					No proficiency
	No proficiency	Slightly proficient	Moderately proficient	Very proficient	Extremely proficient	proficiency
Importing data	☐	☐	☐	☐	☐	☐
Formatting data	☐	☐	☐	☐	☐	☐
Plotting a figure	☐	☐	☐	☐	☐	☐
Creating a for-loop	☐	☐	☐	☐	☐	☐
Creating a function	☐	☐	☐	☐	☐	☐
Using external packages in R	☐	☐	☐	☐	☐	☐
Debugging and troubleshooting	☐	☐	☐	☐	☐	☐
Annotating/documenting code	☐	☐	☐	☐	☐	☐
Data summarization	☐	☐	☐	☐	☐	☐
RMarkdown	☐	☐	☐	☐	☐	☐

How would you rate your proficiency in the following quantitative skills?

	Before this semester					After	
	No proficiency	Slightly proficient	Moderately proficient	Very proficient	Extremely proficient	No proficiency	Slightly proficient
Hypothesis testing and experimental design	☐	☐	☐	☐	☐	☐	☐
Calculating measures of central tendency (e.g., mean, median) or variance	☐	☐	☐	☐	☐	☐	☐
Using statistical tests to compare means (e.g., ANOVA, t-test)	☐	☐	☐	☐	☐	☐	☐
Linear regression analysis	☐	☐	☐	☐	☐	☐	☐
Model output interpretation	☐	☐	☐	☐	☐	☐	☐
Comparing fit among models	☐	☐	☐	☐	☐	☐	☐
Exploratory data analysis	☐	☐	☐	☐	☐	☐	☐

Comfort with R

Answer the following questions about your level of ease in finding external help:

	Not at all comfortable	Slightly comfortable	Moderately comfortable	Very comfortable	Extremely comfortable
How comfortable do you feel with finding help from your classmates with R?	☐	☐	☐	☐	☐
How comfortable do you feel with finding help from the internet (e.g., YouTube videos, StackExchange, etc.) with R?	☐	☐	☐	☐	☐
How comfortable do you feel using official R documentation (e.g., R help files, vignettes)?	☐	☐	☐	☐	☐

In general, when faced with an issue in R, how long do you spend working on it before you give up trying to deal with it?

- ☐ 0 - 5 minutes
- ☐ 5 - 30 minutes
- ☐ 30 minutes - 1 hour
- ☐ 1 - 2 hours
- ☐ 2 - 7 hours
- ☐ 7+ hours

In general, how computer savvy would you consider yourself?

- ☐ Not computer savvy at all
☐ Slightly computer savvy
☐ Moderately computer savvy
☐ Very computer savvy
☐ Extremely computer savvy

Engagement

How much effort was required to accomplish your level of performance in this course?

- ☐ No effort
☐ Low effort
☐ Moderate effort
☐ High effort
☐ Very high effort

In your opinion, in a typical week of class, how engaged were you in the course (e.g., watching/reading required material before class, attending lectures/labs, completing assignments, etc.)?

- ☐ Not engaged
☐ Slightly engaged
☐ Moderately engaged
☐ Very engaged
☐ Extremely engaged

Applicability

How interested are you in taking classes with quantitative content in the future?

	Before this semester					After this semester		
	Not interested	Slightly interested	Moderately interested	Very interested	Extremely interested	Not interested	Slightly interested	Moderately interested
How interested are you in taking classes with quantitative content in the future?	☐	☐	☐	☐	☐	☐	☐	☐

To what degree do you think your R coding skills will be helpful in your future job search?

- ☐ Not at all helpful
- ☐ Slightly helpful
- ☐ Moderately helpful
- ☐ Very helpful
- ☐ Extremely helpful

How interested are you in a career that requires computational skills (i.e. uses R, Python, etc. regularly)?

- ☐ Not interested
- ☐ Slightly interested
- ☐ Moderately interested
- ☐ Very interested
- ☐ Extremely interested

Demographics

What is your first and last name?

How old are you?

How do you describe yourself?

- ☐ Male
- ☐ Female
- ☐ Non-binary
- ☐ Other
- ☐ Prefer not to say

Choose one or more races that you consider yourself to be

- ☐ White or Caucasian
- ☐ Black or African American
- ☐ American Indian/Native American or Alaska Native
- ☐ Asian
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ Other
- ☐ Prefer not to say

Are you of Spanish, Hispanic, or Latino origin?

- ☐ Yes
☐ No

Academic demographics

Which of the following courses are you taking this semester?

- ☐ Computational Problem Solving in Wildlife Ecology Using R (WIS 4934)
☐ Introduction to Wildlife Population Ecology (WIS 4501)
☐ Quantitative Wildlife Ecology (WIS 4601)
☐ Other

What is your major?

What is your current academic standing?

- ☐ Freshman
☐ Sophomore
☐ Junior
☐ Senior
☐ Other
☐ Dual undergraduate/graduate degree program
☐ Graduate degree program

Indicate whether you fall into any of the following categories:

	Yes	No
Transfer student	☐	☐
Nontraditional student	☐	☐
First generation student	☐	☐

Did your high school offer computer science/programming or advanced mathematics classes?

- ☐ Yes
☐ No

☐ I'm not sure

Did you take any computer science/programming or
advanced mathematics classes in high school?

☐ Yes

☐ No

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